

AA8 - Context free grammars

(possíveis soluções – podem estar erradas)

NOTA: contacta-nos se tiveres as soluções corretas accio.feup.dei@gmail.com

1

Consider the following CFG:

$S \rightarrow aS \mid abS \mid A$

$A \rightarrow a \mid ba$

This CFG is:

(-/1 Ponto)

- ☐ Ambiguous and represents a non-regular language.
- ☐ Not ambiguous and represents a non-regular language.
- ☒ Ambiguous and represents a regular language.
- ☐ Not ambiguous and represents a regular language.

2

Consider the following CFG:

$S \rightarrow aaS \mid Sbb \mid A$

$A \rightarrow ab$

The language represented by this CFG can be informally described as: 

(-/1 Ponto)

- ☒ All the strings that have an odd number of 'a's in the beginning and an odd number of 'b's in the end.
- ☐ All the strings that have an even number of 'a's in the beginning and an even number of 'b's in the end.
- ☐ All the strings that have an odd number of 'a's and an odd number of 'b's in any position.
- ☐ All the strings that have an even number of 'a's and an even number of 'b's in any position.

3

Consider the following CFG:

$S \rightarrow 0A \mid 1B$

$A \rightarrow 0S \mid 1C$

$B \rightarrow 0C \mid 1S \mid \epsilon$

$C \rightarrow 0B \mid 1A$

The language represented by this CFG can be informally described as: 
(-/2 Pontos)

- ☐ All the binary strings that have an even number of zeros and an even number of ones.
- ☒ All the binary strings that have an even number of zeros and an odd number of ones.
- ☐ All the binary strings that have an odd number of zeros and an odd number of ones.
- ☐ All the binary strings that have an odd number of zeros and an even number of ones.

4

Consider the following CFG:

$S \rightarrow AaBC$

$A \rightarrow bA \mid a$

$B \rightarrow aB \mid b$

$C \rightarrow AaB \mid \epsilon$

A left-most derivation of the string 'baabaab' is: 
(-/1 Ponto)

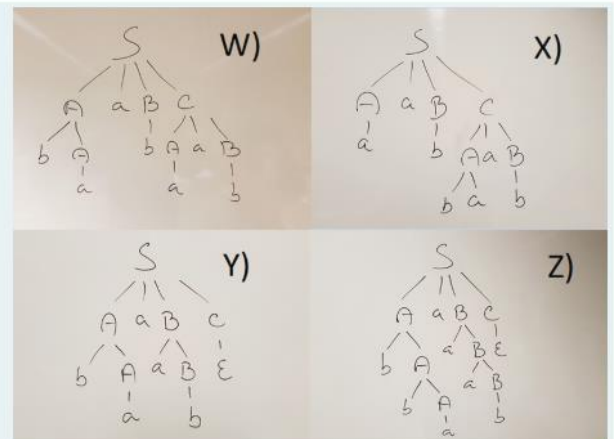
- ☐ $S \rightarrow AaBC \rightarrow AabC \rightarrow bAabC \rightarrow baabC \rightarrow baabAaB \rightarrow baabaaB \rightarrow baabaab$
- ☒ $S \rightarrow AaBC \rightarrow bAaBC \rightarrow baaBC \rightarrow baabC \rightarrow baabAaB \rightarrow baabaaB \rightarrow baabaab$
- ☐ $S \rightarrow AaBC \rightarrow AaBAaBC \rightarrow AaBAaB \rightarrow AaBAab \rightarrow AaBaab \rightarrow Aabaab \rightarrow bAabaab \rightarrow baabaab$
- ☐ $S \rightarrow AaBC \rightarrow AaBAaBC \rightarrow bAaBAaBC \rightarrow baaBAaBC \rightarrow baabAaBC \rightarrow baabbaaBC \rightarrow baabbaabC \rightarrow baabbaab$

5

Which of the following CFGs describes the regular expression $0^*(\epsilon|(1|0))1^*$:
(-/1 Ponto)

- ☐ $S \rightarrow 1S \mid A; A \rightarrow 1 \mid 0 \mid B; B \rightarrow 0B \mid \epsilon$
- ☐ $S \rightarrow 0S \mid A; A \rightarrow 1B \mid 0B \mid B; B \rightarrow 1B \mid \epsilon$
- ☐ $S \rightarrow 0S \mid 1 \mid 0 \mid A; A \rightarrow 1A \mid \epsilon$
- ☒ $S \rightarrow 0S \mid 1A \mid 0A \mid A; A \rightarrow 1 \mid 0 \mid \epsilon$

6



Consider the following CFG and the ASTs in the image:

$S \rightarrow AaBC$
 $A \rightarrow bA \mid a$
 $B \rightarrow aB \mid b$
 $C \rightarrow AaB \mid \epsilon$

Which AST represents the string 'baabaab'?
(-/1 Ponto)

- ☒ W

Consider the following CFG:

$$S \rightarrow aS \mid abS \mid A$$
$$A \rightarrow a \mid ba$$

Also consider the following strings:

X) abba

Y) baab

Z) abaa

Which strings are recognized by the CFG? 
(-/1 Ponto)

- ☐ All strings (X, Y, Z)
- ☒ Strings X and Z
- ☐ Only the string X
- ☐ Strings Y and Z